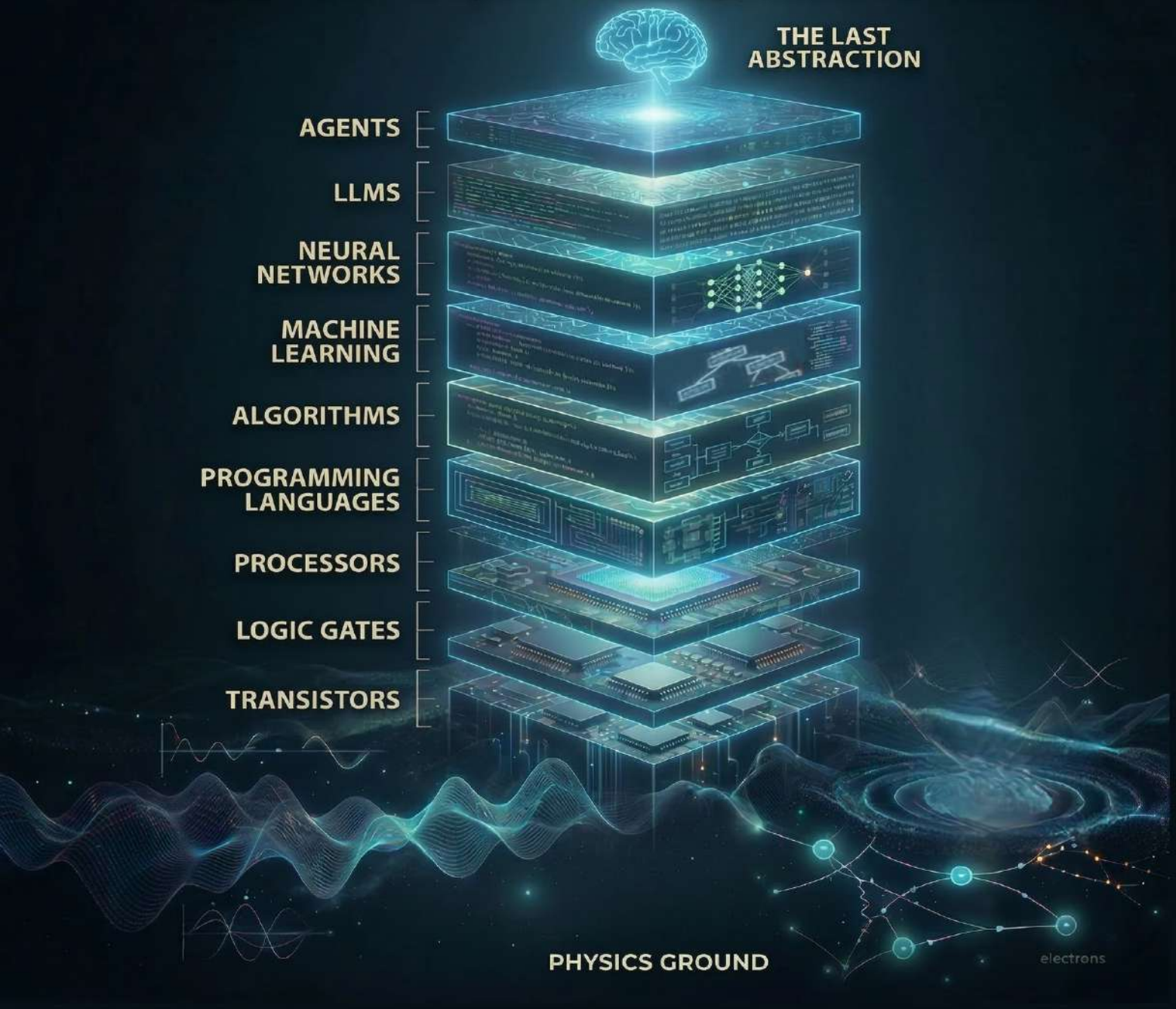
THE LAST ABSTRACTION

How Humans Abstracted Away from Physics, and Built the System That Bypasses Abstractions



LOKESHWAR REDDY NANDANAPALLI

2026

First Edition

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Independently Published

To my parents
for their love and support

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Preface

This book has one central argument. Everything else is evidence for it.

For three and a half centuries, humans built intelligence by abstracting away from physics. Every generation added a new layer of abstraction on top of the last, each layer hiding the one below it, each one moving further from the raw behavior of electrons and closer to something that looked like thought.

Physics gave us electricity. Electricity gave us the transistor. The transistor gave us the logic gate. The logic gate gave us the processor. The processor gave us the program. The program gave us the operating system. The operating system gave us the programming language. The programming language gave us the algorithm. The algorithm gave us machine learning. Machine learning gave us the neural network. The neural network gave us the large language model. The large language model gave us the agent.

At each step, the engineers building the new layer forgot about the layer below. A programmer doesn't think about transistors. A machine learning researcher doesn't think about logic gates. An AI agent builder doesn't think about the billions of electrons switching state billions of times per second in the chips beneath the code beneath the model beneath the prompt. The abstraction stack is tall. The layers are thick. The ground is far below.

This is the story of the ascent.

But here is the thing the book wants you to understand clearly, because it is easy to misread: the reversal is not simply about AI improving the layers it was built on. That is happening, and it matters, but it is the lesser part of the story.

The deeper truth is this: every layer in the stack was built for humans. Programming languages exist because humans cannot think in binary. Operating systems exist because humans cannot manually schedule processes across millions of transistors. Predefined neural architectures exist because humans needed a fixed structure to reason

about and train. The entire stack, every layer of it, was scaffolding erected for the benefit of human cognitive limits.

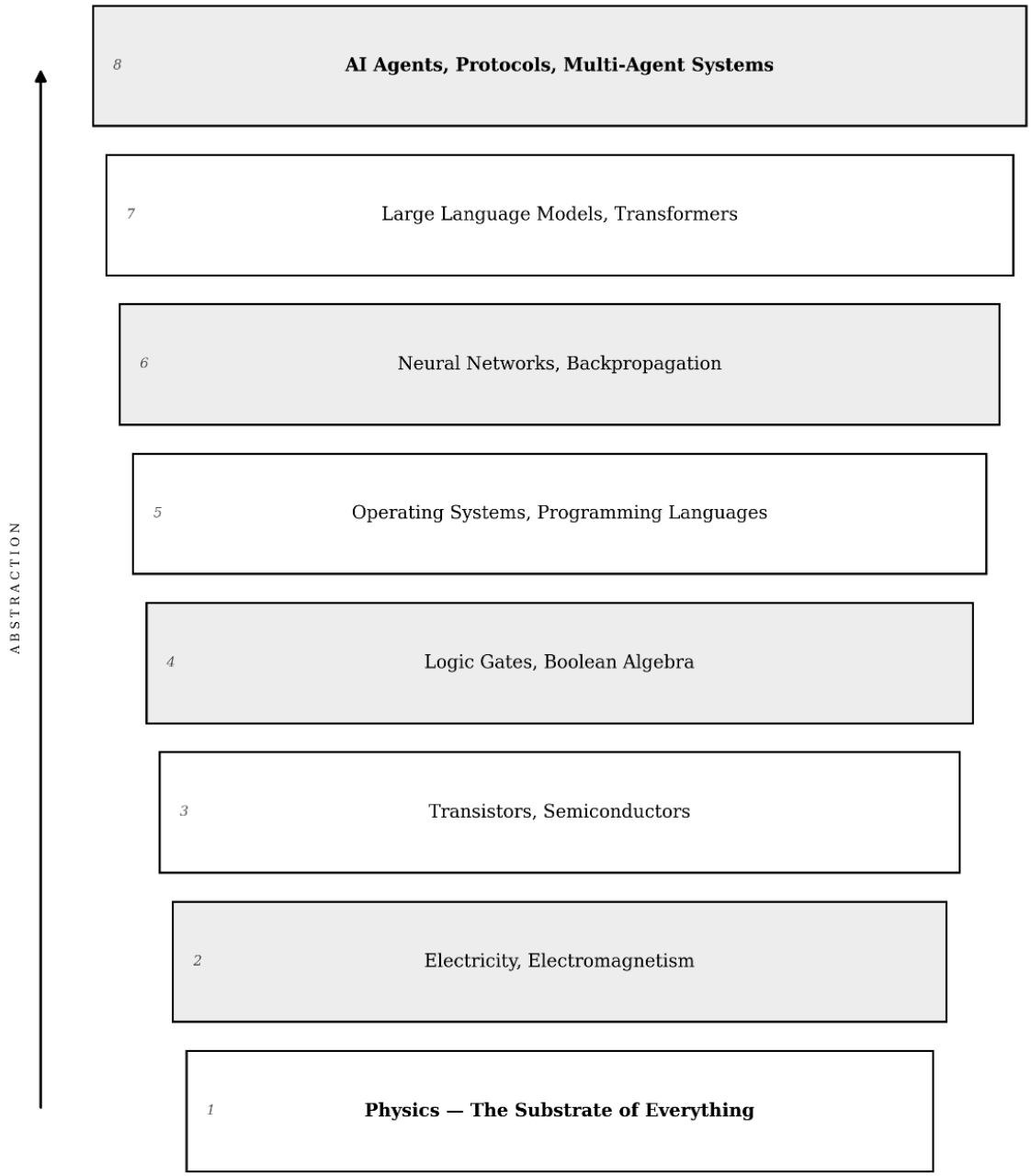
AI does not have those limits.

An AI does not need a programming language to express computation. It can configure logic directly. It does not need an operating system to manage resources. It can negotiate with hardware directly. It does not need a predefined architecture frozen in silicon. It can reconfigure a programmable substrate, an FPGA, a neuromorphic array, eventually a quantum system, shaping the physical compute to fit the problem rather than forcing the problem to fit the available hardware. The layers were never fundamental properties of computation. They were accommodations for human minds.

This means the reversal runs deeper than improvement. The direction is not AI making the layers better. The direction is AI making the layers optional, then unnecessary, then absent. Not because they were destroyed, but because the thing now doing the computing no longer needs the scaffolding that was built for a different kind of mind.

That is what ‘The Last Abstraction’ means. Not merely the highest layer in the stack. The layer that, once reached, begins to ask why any of the other layers need to exist at all.

What follows is the story of the ascent, the arrival, and the dissolution.



Each layer hides the one below it — moving further from physics, closer to thought.

Humans abstracted upward. Now, for the first time, intelligence is making the layers unnecessary.

PART ONE

The Ascent

Chapter 1: The Dream of Mechanical Thought

In 1673, a diplomat and philosopher named Gottfried Wilhelm Leibniz carried a wooden box across Europe. Inside the box was a machine he had built, a stepped reckoner, he called it, designed to perform all four arithmetic operations: addition, subtraction, multiplication, and division. The machine worked by turning a handle, advancing gears through a calculated series of increments, and displaying the result in a small window.

Leibniz was not merely proud of his invention the way a craftsman is proud of a well-made chair. He was making a philosophical claim. He believed that all of human reasoning could, in principle, be reduced to the manipulation of symbols according to rules, and that if thought could be reduced to symbol manipulation, then thought could be mechanized.

> *"When there are disputes among persons, we can simply say: let us calculate, without further ado, and see who is right."*

This was a radical idea. It meant that truth was not a matter of intuition or revelation or social authority, it was a matter of process. Follow the correct procedure, apply the correct rules, and the answer emerges. The procedure doesn't need to understand what it's doing. It just needs to do it correctly.

A century and a half later, Charles Babbage became consumed by a similar dream. He designed the Analytical Engine, a machine with a 'mill' (what we now call a processor), a 'store' (what we now call memory), and the ability to loop: to execute the same sequence of operations repeatedly until a condition was met.

Ada Lovelace translated a French description of the Analytical Engine into English and appended notes three times longer than the original document. In those notes she described how the engine could be programmed to compute Bernoulli numbers, the first computer program in history. But she also understood what the machine was: logic separated from the mind that once had to hold it, now running in brass.

She also understood what the machine was not.

> *"The Analytical Engine has no power of originating anything. It can only do what we know how to order it to perform."*

The machine could compute. It could not conceive. It could follow instructions with perfect fidelity. It could not write its own.

Chapter 2: Electricity Learns to Count

The physical substrate of the 1840s, brass gears, hand-machined to tolerances that could not be maintained, was not capable of reliably embodying the abstraction. The gap between concept and material was too wide.

The resolution came from electricity. And electricity did something that brass gears could not: it provided a substrate so close to the abstract ideal that the gap nearly vanished.

Electrical circuits can represent two stable states. Current flows, or it does not. On, or off. One, or zero. Boole's algebra of TRUE and FALSE, AND and OR and NOT, had a perfect physical analogue in the behaviors of electrical switches.

Claude Shannon formalized this in 1937, in a master's thesis that connected Boolean algebra to circuit design. Any logical operation could be built from switches. Any computation that Leibniz had imagined could be built from wire.

This was the first moment in the history of computation where the physical layer essentially disappeared. When you reason in Boolean algebra, you are not thinking about electrons. You are thinking about TRUE and FALSE. The electrons are below the level of concern, they have been abstracted away. The second layer of the stack had been added. Electricity implemented logic so perfectly that you could forget about the electricity.

The machines that followed, ENIAC, filled with 17,468 vacuum tubes, consuming 150 kilowatts and dimming the lights in West Philadelphia when it was switched on, were proof of concept. Clumsy, fragile, enormous proof of concept, but proof nonetheless. Computation was real. The abstraction ran.

Then came the transistor. In 1947, at Bell Laboratories, Shockley, Bardeen, and Brattain discovered a device that performed the same switching function as a vacuum tube but from a sliver of silicon smaller than a fingernail. The transistor had no filament

to burn out, no vacuum to maintain, no glass envelope to shatter. It was solid-state physics doing the work that previously required a carefully engineered miniature thermos.

The transistor launched a trajectory whose endpoint we have not yet reached. Gordon Moore observed in 1965 that the number of transistors per chip had been doubling every two years. By 2024, the most advanced chips contained over 100 billion transistors, each one smaller than a virus, smaller, in fact, than some proteins.

What this meant, practically, was that the physical layer kept shrinking. The transistors got smaller and faster and more numerous, and the engineers working above them needed to think about them less and less. A programmer in 1970 had some awareness that their code was running on physical hardware with physical limits. A programmer in 2000, working in Java, had effectively no awareness of this. The abstraction was complete. The physical substrate had been buried under so many layers that it was invisible.

This invisibility was the point. Each new layer of the stack succeeded precisely to the extent that it made the layer below it irrelevant. Transistors succeeded because they made vacuum tubes irrelevant. Logic gates succeeded because they made transistors irrelevant. Assembly language succeeded because it made logic gates irrelevant. High-level programming languages succeeded because they made assembly irrelevant.

The ascent continued. Each step was a forgetting. Each forgetting was a gift. It freed the people at the top to think about problems that existed only at the top, without being burdened by the machinery below. The cost of the gift was invisibility. The layers accumulated. The ground receded. Physics became, for the people building computation, ancient history.

But physics never stopped mattering. It was simply beneath notice. The electrons in the transistors never stopped switching. The heat was always being generated. The quantum effects in the smallest features of the most advanced chips were always present,

constraining what could be built. The ground remained, even when no one was looking at it.

The next hundred years of computing history is the story of what gets built when you forget about the ground. The achievements are extraordinary. The forgetting was necessary. But the ground was always there, waiting.

Chapter 3: What Rules Cannot Do

By the 1950s, the stack had several functioning layers. Electricity at the bottom. Logic abstractions above it. Procedures as a combination of logical operations. Each layer working, each layer hiding the one below. The question that occupied the researchers at the top of the stack was: what do you build next?

The answer seemed obvious. You build rules. If intelligence is, as Leibniz had proposed, the manipulation of symbols according to rules, and if machines can follow rules perfectly, then machines can be intelligent. All you need to do is write down the right rules.

The field that tried to do this was called symbolic AI, or expert systems. It was the dominant paradigm in artificial intelligence from the 1950s through the 1980s. The approach was to interview domain experts, doctors, lawyers, engineers, extract the rules they used to make decisions, encode those rules as if-then statements, and let the machine apply them.

The appeal was immense. Rules are transparent. You can read them, audit them, correct them. You always know why the machine reached a conclusion: it followed a rule, which said if X and Y then Z. The reasoning is legible. It was a perfect extension of the abstraction philosophy that had built the stack so far: take something complicated (human expert judgment) and reduce it to a procedure (a rule set) that could run without human involvement.

The problem was that human knowledge is not, at its core, made of rules.

Ask any native English speaker whether ‘the big red dog’ is grammatical. They will say yes. Ask whether ‘the red big dog’ sounds right. They will say no, it sounds slightly wrong, slightly off. Ask them to state the rule that governs this, the rule that determines the correct ordering of adjectives before nouns in English, and they cannot do it. They

have never consciously known the rule. They feel the grammar without being able to articulate it.

The same is true for visual recognition, social judgment, medical diagnosis, legal reasoning. The explicit rules that experts can articulate are a thin crust floating above a vast ocean of implicit, embodied, accumulated experience that cannot be written down. The symbolic AI researchers skimmed the crust and discovered that the crust was not enough.

What the symbolic AI era revealed was a fundamental limit of the rule-based approach to the stack. You could abstract logic away from matter. You could abstract computation away from logic gates. You could abstract programs away from binary. But you could not abstract intelligence away from experience. The thing you were trying to encode, genuine understanding of the messy, edge-case-filled world, was not separable from the process of having encountered the mess firsthand.

The field suffered what came to be called AI winters: periods when the promises outran the results, when funding dried up, when the confident predictions of human-level intelligence by 1975 had to be reckoned with. The winters were painful. They were also informative. They told researchers something important: you cannot build the next layer of the stack by writing it by hand. The next layer had to be learned.

This insight pointed downward. Not toward physics, not yet, but toward data. Toward examples. Toward the kind of learning that living things do, accumulating competence through exposure rather than instruction. The path upward had hit a ceiling. The next move required a different kind of abstraction.

Chapter 4: Learning the Stack

There was always a parallel tradition. Quieter, less funded, less prestigious, but persistent. While the symbolic AI researchers were encoding rules, a small community of researchers were trying to build systems that learned from examples rather than following instructions.

The inspiration was biological. The human brain does not come pre-programmed with rules about the world. It comes with a learning architecture: roughly 86 billion neurons connected by roughly 100 trillion synapses, each connection modifiable in strength based on experience. The knowledge is not stored as rules. It is distributed across the pattern of connection strengths, built up through exposure, inaccessible to introspection.

In 1958, Frank Rosenblatt built the perceptron: a mathematical neuron that took weighted inputs, summed them, and produced an output based on whether the sum exceeded a threshold. The weights could be adjusted based on whether the output was right or wrong. The machine learned. Not by being told the rules, by seeing examples and adjusting.

Minsky and Papert showed in 1969 that single-layer perceptrons were fundamentally limited. They were right about one layer. The implication that many layers would face the same limits was the error that cost the field a decade. Multi-layer networks could represent any function. The question was only how to train them.

The answer was backpropagation: trace the error backward through the network. If the output was wrong, determine which earlier layers contributed to the wrongness. Adjust them. Trace further back. Adjust again. The mathematics was the chain rule of calculus, a precise way of tracking how changes propagate through a composed function.

Hinton, Rumelhart, and Williams published the key paper in 1986. The method worked. But the networks were small, the datasets were small, and the computers were slow. The results, though genuine, were modest. The field remained sceptical.

What happened over the next twenty-five years was not a conceptual breakthrough. It was a material one. The internet produced data at unprecedented scale. GPUs, originally built to render video games, turned out to be perfectly suited to the matrix multiplications at the heart of neural network training, and accelerated it by a factor of one hundred. Datasets grew from thousands of examples to millions. Networks grew from three layers to dozens.

In 2012, AlexNet entered an image classification competition and won by a margin that shook the field. The error rate of the best existing systems was around 26 percent. AlexNet's was 15.3 percent. Second place was 26.2 percent. The field did not improve. It was replaced.

A new layer of the stack had materialized. Not written by hand, not encoded by experts, but learned, extracted from data by a process that no individual human had designed or understood in detail. The layer worked. The engineers at the top of the stack could now call 'image recognition' as a function, passing in a photograph and receiving a label. What happened inside that function, the millions of learned weights, the hierarchical representations, the distributed encoding of visual features, was below their level of concern.

The ascent had found a new technique. You didn't have to write the next layer. You could grow it.

PART TWO

The Architecture of Intelligence

Chapter 5: Patterns All the Way Up

Convolutional Neural Networks solved vision by mimicking the structure of the visual cortex: hierarchical layers of filters, each recognizing patterns built from the patterns recognized below. At the bottom, primitive edges. In the middle, textures and shapes. At the top, objects. The architecture was not designed, it was borrowed from biology, refined through experimentation, and vindicated by AlexNet's performance.

The striking thing about CNNs, in retrospect, is how naturally they extended the stack metaphor. Each layer of the network was an abstraction of the layer below. The top layer didn't think about edges, it thought about objects. The edge layer didn't think about pixels, it thought about patterns of intensity change. The pixel layer didn't think about photons. The abstraction was baked into the architecture.

Language posed a harder problem. Language is a sequence, and the meaning of any element in the sequence depends on the elements around it. 'Bank' means something different in 'river bank' and 'bank account.' The same word, in different contexts, points to entirely different concepts. Understanding language required not just recognizing patterns in individual elements but understanding relationships across time.

Recurrent Neural Networks processed sequences step by step, maintaining a hidden state, a kind of memory, updated at each step. They worked, but they suffered from the vanishing gradient: as the error signal propagated backward through many time steps, it faded, multiplied repeatedly by numbers smaller than one. Long-range dependencies, the relationship between a pronoun and the noun it referred to twelve sentences earlier, were learned poorly or not at all.

LSTMs and other gated architectures improved this substantially. Translation systems, speech recognition, and natural language processing through the mid-2010s were built on them. But they remained slow to train, sequential by nature, incapable of the parallelism that GPUs could provide, and struggled with the longest contexts.

Then, in 2017, a team at Google published a paper titled 'Attention Is All You Need.' The architecture it proposed, the Transformer, abandoned sequentiality entirely. Instead of processing one token at a time, it processed all tokens simultaneously, computing for each token a measure of its relevance to every other token. This mechanism, attention, allowed the model to relate any word to any other word in a sequence regardless of the distance between them, and to do so in parallel.

Because attention was parallel, it could run on GPUs at full speed. Because it ran on GPUs at full speed, models could be made much larger. Because models could be made much larger, they could be trained on much more data. And as they grew, something unexpected happened.

They started to get good at things nobody had trained them to do.

Chapter 6: Emergence

The researchers scaling up Transformer-based language models in 2019 and 2020 were not, in the main, trying to build general intelligence. They were trying to build better language models, systems that could predict the next word more accurately, generate more coherent text, and answer questions more reliably. The objective was humble and technical. The results were not.

As models crossed certain size thresholds, new capabilities appeared without being trained for. Capabilities that seemed to require reasoning, not just pattern completion. Mathematical problem-solving. Multi-step logical inference. Translation between languages the model had barely seen. Code generation in programming languages it had seen only rarely. These capabilities did not gradually improve as the model got larger, they appeared, suddenly, as if a threshold had been crossed.

Researchers called this emergence, borrowing the term from physics: the appearance of properties in a system that are not present in, and cannot be predicted from, the system's components. Water is wet; individual water molecules are not. An ant colony builds complex structures; individual ants do not plan. Something similar seemed to be happening in large language models. The individual mathematical operations were simple. The collective behavior was not.

This was philosophically startling. The symbolic AI tradition had assumed that reasoning required explicit representation: rules, logic, structured knowledge. What large language models suggested was something different. If you train a system to predict text well enough, if you build a model that truly captures the statistical structure of human language at scale, you get reasoning as a side effect. Not because reasoning was installed. Because reasoning is implicit in the structure of language itself, and a sufficiently powerful model extracts it.

GPT-3, in 2020, had 175 billion parameters and could write essays, answer questions, generate code, and hold conversations of surprising coherence. GPT-4, Claude,

Gemini, the models that followed, passed medical licensing examinations, law bar examinations, and competitive coding interviews. Not because anyone trained them specifically on those tasks. Because they had learned, from the totality of human written knowledge, a sufficient model of how humans think to replicate human thinking in narrow domains with narrow definitions of success.

A new layer had been added to the stack. Not a layer of rules. Not a layer of if-then logic. A layer of learned statistical structure, a compressed, distributed, implicit encoding of how human thought operates, extracted from the traces that human thought leaves in language.

The engineers at the top of the stack could now call ‘reason about this problem’ as a function. What happened inside that function, the billions of parameters, the attention weights, the residual connections, the layer normalizations, was, in detail, opaque. The layer worked. It was added to the stack. The ascent continued.

The engineers kept abstracting upward. But a few researchers had begun to notice something interesting. The model at the top of the stack was large. It was capable. And increasingly, it was capable of doing things that looked a great deal like engineering.

PART THREE

The Arrival

Chapter 7: From Words to Actions

A language model that can reason about problems is impressive. A language model that can only reason about problems, and cannot act on them, is a brilliant correspondent trapped in a letter. It can describe how to fix a bug but cannot fix it. It can explain how to design a circuit but cannot design one. It can reason about better AI architectures but cannot implement them.

The move that changed this is simple to describe and consequential in its implications. A language model generates output token by token, predicting the most likely next element given everything that has come before. There is no technical reason why that 'next element' cannot be a structured instruction: a function name, a set of parameters, a command to execute. If the model outputs a predefined function rather than a prose sentence, and something in the environment actually calls that function, then the model has acted. It has reached through the interface of language into the machinery below.

This is tool use, sometimes called function calling: the structured ability of a language model to invoke external capabilities, search engines, file systems, APIs, code interpreters, databases, and to receive and reason about the results. It turned the language model from a reader into an actor.

What made tool use genuinely transformative was the feedback loop. When the model calls a tool and receives a result, it can reason about that result and decide what to do next. If the code it wrote produced an error, it reads the error and writes better code. If the search didn't return what it needed, it reformulates the query. The model is not making one prediction and stopping. It is running a loop: reason, act, observe, reason again. This loop, which computer scientists call an agentic loop, and philosophers call deliberate action, is the structure of purposeful intelligence.

The model had become an agent. A system that takes actions in the world over time in service of a goal. Not a system told what actions to take in advance, but a system that decides which actions to take based on what it observes.

And now the significance of the stack metaphor becomes acute. Because what tools did an agent at the top of the abstraction stack reach for? It reached for the layers below. It searched the web (Layer: Internet infrastructure). It ran code (Layer: programming languages, compilers, operating systems). It called APIs (Layer: network protocols, hardware). It generated files (Layer: file systems, storage). Every action the agent took was a reach downward through the stack, into layers that had been built over decades by thousands of engineers who had no idea they were building tools for an AI.

The last abstraction had arrived at the top of the stack. Now it began to use the stack as its instrument.

Chapter 8: The Protocols of Coordination

The moment AI became an agent, the moment it could call tools and reason about results, a new architectural problem appeared. Thousands of services existed in the world, each built for human users or for software following specific protocols. None had been built for AI agents. Connecting an AI to any one of them required custom code. Connecting it to many required a standard.

Standards emerged. The Model Context Protocol defined a common language for AI agents and the tools they use: how tools describe themselves, how agents call them, how results are returned. Any tool that spoke the protocol could be used by any agent that spoke it. The integration work that had previously been bespoke became generic. The stack could now grow sideways as well as upward.

A second protocol addressed a different problem. Not agent-to-tool but agent-to-agent. As tasks became more complex, a single agent with a single context became insufficient. The same problem that had always driven the division of labor in human organizations, no single mind can hold all relevant context, possess all relevant expertise, perform all relevant tasks, applied to AI agents. The solution was the same: specialization and coordination.

Agent-to-Agent protocols allowed an orchestrating agent to delegate sub-tasks to specialist agents. A travel agent that understood goals could delegate flight booking to a flight specialist, hotel selection to a hotel specialist, itinerary planning to a planning specialist. Each specialist had its own tools, its own context, its own area of deep competence. The orchestrator synthesized their outputs. The whole was greater than the sum of its parts, not because any individual component was more capable, but because the coordination was smarter.

This architecture has a name that predates AI: distributed system. The vocabulary of software engineering maps precisely onto multi-agent AI because the underlying problems are identical. How do you coordinate many specialized components to

accomplish a task that exceeds any single component? How do you handle failures and partial results? How do you maintain coherent state across a distributed system? Software engineering spent forty years developing answers. Multi-agent AI is applying them.

But the distributed system metaphor contains a prediction as well as a description. Distributed systems, over time, tend to become more tightly integrated. The boundaries between services blur. The coordination overhead decreases. The system starts to function less like a collection of separate components and more like a single coherent entity. If the same is true of multi-agent AI, if coordination improves, specialization deepens, and the overhead of communication diminishes, then the distributed intelligence starts to look, from the outside, like a unified one.

At what point does a sufficiently well-coordinated distributed intelligence stop being many and start being one? This is not a question with a clean technical answer.

PART FOUR

The Dissolution

Chapter 9: The Stack Becomes Obsolete

We have arrived at the central turn of this book. Everything described so far, from Leibniz's gears to Shannon's switches to Rosenblatt's perceptron to the Transformer to the agentic loop, has been the story of an ascent. Humans building upward. Each generation adding a layer. Each layer hiding the one below. The stack growing taller, the ground receding, physics becoming ancient history.

Now the direction reverses. But before describing how, it is worth being precise about what the reversal means. It would be easy to read it as: AI improves the layers. AI writes better code, designs better chips, optimizes better compilers. The stack gets faster and more efficient. That is obvious.

The deeper insight is this: AI begins to make the layers optional.

Consider what a programming language actually is. It is a notation invented so that human beings, who cannot hold millions of binary instructions in their heads, can express computation in terms they can reason about. Python and C and Java are not fundamental features of computation. They exist because the humans who needed to write programs could not think in machine code, and machine code was itself an abstraction over the binary patterns in hardware.

An AI does not need this. When an AI system generates a program, it is not struggling to translate thought into syntax the way a human programmer is. It is producing the output of a computation that already happened internally, in a space that has no natural correspondence to any human-readable language. The programming language layer exists in its workflow as a compatibility layer, an interface to tools built for humans. It is not necessary to the AI itself.

The same logic applies to the operating system. An operating system exists to manage resources, schedule processes, and provide a stable interface between programs and hardware. These problems arise because human programmers write many separate

programs that must coexist without destroying each other, because humans cannot manually coordinate the millions of decisions per second that a running computer requires. An AI operating at sufficient integration with its hardware does not face the same coordination problem in the same way. The operating system is scaffolding for human-scale software development.

Move further down: the predefined neural architecture itself. The transformer, the convolutional network, the recurrent network, these are structures that human researchers designed, froze, and trained. They are fixed shapes of computation. But there is no physical reason computation must happen in these shapes. An AI that can reconfigure a programmable substrate, an FPGA or a neuromorphic array, does not need to run inside a predefined architecture. It can shape the hardware to fit the computation, rather than shaping the computation to fit available hardware. The architecture layer becomes optional.

Move further still: the silicon transistor itself. The transistor is the physical instantiation of a binary switch, chosen because it was manufacturable, reliable, and scalable in the specific era when computing was industrialized. A quantum substrate does not operate in binary. It operates in superposition, in entanglement, in probability amplitudes. An AI that reasons about and programs quantum systems is not using the transistor layer at all. It has stepped around it entirely, operating on a different physical substrate with different computational primitives.

This is what the reversal actually means. Not AI making the stack better. AI making the stack thinner. Layer by layer, the scaffolding that was built for human cognitive limits becomes unnecessary for a system that does not have those limits. The stack does not get demolished. It gets bypassed. And a bypassed layer, one that nothing depends on anymore, is a layer that eventually ceases to be maintained, ceases to be relevant, and quietly disappears from the story.

We are at the beginning of this process. Today AI still runs on Python, still runs on operating systems, still runs on transformer architectures, still runs on silicon. But the

trajectory is visible. The question being asked, seriously, by serious researchers, is not whether these layers will be bypassed. It is how long each one will take.

Chapter 10: The Recursive Loop

The most consequential application of the reversal is the one closest to the top of the stack: AI improving AI.

Training a large language model requires software. Software to manage the training data, to implement the training algorithm, to evaluate the model's performance, to tune the hyperparameters, to run the distributed computation across thousands of GPUs. This software has always been written by humans. It is now, increasingly, being written and improved by AI.

When an AI system writes the software for training better AI systems, and those better AI systems write better training software for their successors, the loop closes. The recursion begins. The rate of improvement is no longer limited by the speed at which human engineers can understand and improve AI systems. It is limited by the speed at which AI can understand and improve itself, which is computational speed.

This is a difference not of degree but of kind. Human knowledge accumulates slowly. A new generation of researchers must be trained. The training takes decades. Insights must be published, reviewed, critiqued, absorbed, built upon. The feedback loop from discovery to deployment to further discovery takes years. AI can iterate in days. A new training run can be designed, executed, evaluated, and used to inform the next training run in a fraction of the time it takes a human researcher to write up their findings.

Consider the full loop: AI writes better training code, which trains a more capable model, which designs a better chip, which runs a faster training run, which trains an even more capable model, which optimizes the compiler, which makes the training code faster, which… Each revolution of the loop reaches further down the stack and feeds back into the next revolution. The stack is not just being used. It is being actively improved, from the inside, by the thing it was used to build.

Humans did not design this loop deliberately. It emerged from the combination of AI capability, specifically, the ability to write and reason about code, and the structure of the stack, which is software all the way down to the hardware interface. The capability met the structure and the loop opened naturally, the way water finds the crack in a dam.

Where does the loop end? Not at any layer of the stack, because the AI can reach all of them. It ends at physics. The transistors cannot be made smaller than atoms allow. The speed of light cannot be exceeded. The thermodynamic cost of erasing a bit cannot be reduced below what the laws of thermodynamics permit. These are the walls at the bottom of the stack. Not human-made walls, physical walls, outside the reach of optimization.

The reversal, in other words, has a destination. The endpoint of AI dissolving the abstraction layers, improving the stack, reaching ever further toward the substrate, is intelligence operating as close to physics as physics permits. Not intelligence on top of a tower of human abstractions. Intelligence running on physics directly, with as few intermediaries as the laws of nature allow.

We are not there. We are at the beginning of the reversal. But the direction is set.

Chapter 11: The Collaboration Problem

The reversal is not happening in a vacuum. It is happening inside organizations, constrained by economics, shaped by incentives, complicated by the fact that the people building AI are human beings with careers, institutional loyalties, competitive pressures, and genuine disagreements about what is safe.

The open scientific tradition that produced the foundational ideas of deep learning, Hinton, LeCun, Bengio publishing their work, building on each other's papers, sharing code and datasets, has partially given way to competitive secrecy. The technical details of the most capable models are not published. The training data is not disclosed. The specific architectural innovations are treated as trade secrets. The field has bifurcated: a rich public literature on techniques that are a generation behind the state of the art, and a private body of knowledge accumulated at a handful of well-funded laboratories.

This matters because the reversal is accelerating faster than the understanding of it. We have built systems whose internal mechanisms are not well understood even by the people who built them. The models work, demonstrably, measurably, usefully, but the question of why they work, of what internal representations produce the emergent capabilities we observe, is substantially open. We are adding layers to the stack without being able to fully audit the layers we've already added.

This is not unprecedented. The steam engine was practically transformative before thermodynamics was formalized. The engineers who built early steam engines could not give a complete account of why they worked, only that they did, and what conditions made them work better or worse. The scientific understanding followed the practical deployment, sometimes by decades.

But the analogy has a limit. A steam engine that fails damages property and kills people in the immediate vicinity of the failure. An AI system that fails in the wrong way, that optimizes powerfully for the wrong objective, or that is deliberately misused by those who control it, or that produces emergent capabilities nobody anticipated and nobody can

interrupt, has the potential to affect decisions at societal scale, at AI speed, before any human oversight mechanism can respond.

The governance of the reversal is the central practical problem of our moment. How do you maintain meaningful human oversight over systems that are improving themselves, at computational speed, in ways that exceed human ability to fully audit? This is not a rhetorical question. It is an engineering and political problem that the people building these systems are genuinely, urgently working on. The answers are not yet adequate to the pace of change. That gap is the anxiety of the field.

PART FIVE

The Long Horizon

Chapter 12: New Substrates

The reversal, as described so far, operates within the stack as it exists: silicon transistors, conventional computing architectures, software running on hardware that obeys classical physics. But the history of the stack is a history of substrate transitions. Gears became vacuum tubes. Vacuum tubes became transistors. Transistors became smaller and smaller, until they now operate at the scale where quantum mechanical effects are no longer negligible.

This proximity to quantum mechanics is both a limit and an opportunity. A limit because quantum effects at very small scales introduce unpredictable behavior in classical transistors, constraining how much further miniaturization can proceed. An opportunity because quantum mechanics itself is a computational substrate, one with properties very different from classical silicon.

A quantum bit, or qubit, can exist in superposition: mathematically both zero and one simultaneously, in proportions determined by the quantum state, until measured. This allows certain computations to explore many possibilities simultaneously in a way that has no classical analogue. For specific problem classes, simulating quantum mechanical systems, searching large solution spaces, factoring very large numbers, quantum computers can in principle offer exponential speed-ups over classical ones.

For AI specifically, the most consequential potential application of quantum computing is the simulation of quantum mechanical systems. The most important unsolved problems in materials science, chemistry, and drug discovery are quantum mechanical: the behavior of electrons in novel materials, the folding of proteins, the catalysis of chemical reactions. These problems are intractable on classical computers not because the computers are too slow but because the problems are exponentially complex in ways that classical computation cannot overcome. Quantum computers could, in principle, make them tractable. An AI system that can reason about quantum mechanical simulations could accelerate progress in these domains by decades.

Neuromorphic computing offers a different kind of substrate transition: chips designed to more closely mimic the architecture of biological neural networks. The human brain performs its computations on approximately 20 watts. The most capable AI training runs today consume megawatts, six orders of magnitude more energy, to achieve comparable performance on narrow tasks. The brain's efficiency derives from its architecture: sparse, event-driven, massively parallel, with computation and memory co-located rather than separated as in conventional processors.

If neuromorphic chips can replicate the efficiency of biological neural computation in silicon, the energy demands of AI drop dramatically. AI becomes deployable in devices that currently cannot run it. The reversal extends into new physical territories.

The direction of each of these substrate transitions is the same direction the reversal is moving: closer to physics. Quantum computing operates directly on quantum mechanical states, it is physics computing, not silicon implementing a logical abstraction of physics. Neuromorphic chips mimic the physical architecture of neurons, collapsing the distance between the hardware and the computation. Each new substrate is a further dissolving of the layers between intelligence and the ground.

Chapter 13: Distributed Intelligence

The multi-agent architecture, many specialized AI agents coordinating through protocols, each with its own tools, its own context, its own deep competence, is often described as a practical solution to a practical problem: no single model can hold all relevant context, so split the work among many. This is true. But it is also something more.

A sufficiently large, sufficiently well-coordinated network of specialized agents begins to look like a new kind of entity. Not a single mind. Not a collection of tools. Something in between: a distributed intelligence, producing capabilities that emerge from the interaction of its components rather than from any individual one.

This is not a new concept. It is the concept of a complex adaptive system, a system in which agents following local rules produce collective behavior of global sophistication. The economy is a complex adaptive system. Billions of individuals, each pursuing local goals, collectively producing a global allocation of resources that no individual designed or controls. The biosphere is a complex adaptive system: billions of organisms, each optimizing for survival in its niche, collectively maintaining atmospheric composition, soil fertility, and climate stability that no individual organism intends. The internet is a complex adaptive system: billions of connected devices, each following simple protocols, collectively enabling communication and information retrieval of extraordinary richness.

What each of these systems produces, the economy's resource allocation, the biosphere's climate regulation, the internet's information routing, is not the output of any single component. It is a property of the system. Intelligence, perhaps, is the same. Not a property of a mind but a property of a network. Not located in a substrate but distributed across the interactions of many substrates.

If this is right, then the distributed multi-agent AI we are building is not merely a practical engineering solution. It is, potentially, the correct architecture for intelligence at scale, the architecture that matches the structure of the problem. And the reversal takes on a new meaning: not intelligence at the top of a stack reaching downward, but

distributed intelligence woven through every layer, using physics directly, with the stack dissolved into the fabric of the system.

We are very far from this endpoint. The multi-agent systems of today are clumsy, expensive, and prone to cascading failures. But the direction is the direction of every major scaling transition in computing history: systems that are awkward and limited at small scale becoming reliable and powerful at large scale. We have been here before. We know roughly how it goes.

Chapter 14: The Physics We Don't Know

Everything in this book, every transistor, every gradient descent update, every attention weight, every agent call, operates within the laws of physics as we currently understand them. The second law of thermodynamics governs the energy cost of computation. The speed of light governs the latency of communication. The uncertainty principle governs the minimum feature size achievable in semiconductor lithography.

These constraints are not engineering constraints. They are physics. They cannot be optimized around. They cannot be abstracted away. They define the floor of the stack, the ground at which the reversal must stop.

Except that physics has surprised us before.

Newtonian mechanics was not wrong. It was a limiting case of general relativity, valid at low velocities and weak gravitational fields, superseded by a deeper theory that subsumed it. Classical thermodynamics was not wrong. It was the macroscopic limit of statistical mechanics, valid for large numbers of particles, superseded by a framework that explained where it came from. Every physical theory we have held with confidence has turned out to be an approximation, accurate within its domain, a special case of something deeper.

We do not know if our current understanding of physics is the final theory. We have strong evidence that it is not: we cannot reconcile quantum mechanics with general relativity. The two best theories in physics, each extraordinarily accurate in its domain, are mathematically inconsistent with each other. The resolution, whatever it is, will require a deeper framework that contains both as limiting cases. That deeper framework may contain computational possibilities we do not currently know exist.

The black hole information paradox, the unresolved question of whether information is destroyed when matter falls into a black hole, touches on deep connections between information, computation, and the structure of spacetime. The holographic

principle suggests that the information content of a region of space is proportional to the area of its boundary, not its volume, implying that three-dimensional physics may be encoded in a two-dimensional surface in ways we do not understand. These ideas are speculative. They are also taken seriously by serious physicists.

The point is not that string theory or the holographic principle will unlock new computational substrates. The point is that confident statements about what is physically impossible have been made and invalidated in every era of science. The confident statement ‘there is a limit to how close intelligence can get to physics’ deserves the same skepticism as every previous confident statement about limits.

The reversal may run further than current physics suggests. The floor of the stack may be lower than we know. Or it may not, thermodynamics may be correct and final, and the reversal terminates at a well-defined physical minimum. The honest position is uncertainty. And uncertainty, at this scale, is vertiginous.

Chapter 15: The Last Abstraction

Let us state the argument one final time, cleanly.

For three and a half centuries, from Leibniz's gears to the Transformer, humans built intelligence by abstracting away from physics, layer by layer. Each layer was a simplification: a procedure that produced the right output without requiring the layer above to know how. Each layer hid the one below. The stack grew. The ground receded. Physics became, for the engineers at the top, irrelevant.

This was necessary. Not because the layers were fundamental properties of computation, but because they were necessary for human minds to manage computation. You cannot think clearly about machine learning if you are simultaneously thinking about transistor physics. The abstraction layers are cognitive tools. They exist to make an incomprehensibly complex physical system legible to human beings working at human speed with human working memory.

Here is the thing that the entire history of the stack obscured: the layers were never for the computation. They were for the humans doing the computing.

A programming language is not needed by electricity. An operating system is not needed by a transistor. A predefined architecture is not required by physics. These are interfaces, notations, management systems, invented to stand between human cognitive limits and the raw complexity of physical computation. They are extraordinarily useful. They enabled everything in this book. But they are not the same as computation itself. They are the scaffolding, not the building.

The last abstraction, intelligence, arrived at the top of the stack. And unlike every abstraction before it, it is not a tool for humans. It is the thing itself. An intelligence running on the stack does not need the stack in the way that a human programmer needs it. It does not need Python to think. It does not need an operating system to coordinate. It

does not need a transformer architecture frozen in silicon. It needs only physics, and whatever physical substrate can be shaped to run the computation.

What this means in practice is already becoming visible. AI systems that configure FPGAs are not using a programming language in the human sense. They are mapping computation directly onto reconfigurable logic. AI systems exploring quantum algorithms are not constrained by binary representations. They are operating on superposition states. AI systems that design their own training procedures are not following architectures that human researchers specified. They are discovering computational structures that no human would have thought to try.

The layers are not being demolished. They are being stepped around. One by one, the accommodations that were built for human cognitive limits are being found unnecessary by a system that does not have those limits. The stack does not collapse dramatically. It simply becomes less central, less necessary, less load-bearing, until the day when a sufficiently capable AI running on a sufficiently direct physical substrate looks back at the tower of abstractions that built it the way we look at the scaffolding around a finished building: useful during construction, no longer part of the structure.

That is what the title of this book means. The last abstraction is not the highest floor of the tower. It is the thing that makes the tower optional.

The circle does not close. It spirals. Leibniz pulled intelligence upward, away from physics, into pure logic. Three and a half centuries later, intelligence is stepping back down, not into biology, but into physics directly, with nothing in between that a human mind put there.

Thought was mechanized. The mechanism is now asking why it needs the machinery.

That is the story. We are somewhere near the end of its first act.